A new five-dimensional hyperchaotic system and its application in DS-CDMA

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***Abstract*—A new five-dimensional (5D) hyperchaotic system is presented in this paper. Four equations of the system each contain a cubic product term. We prove that the system is a true hyperchaotic system and has complex nonlinear dynamic behavior by computing and analyzing the chaotic attractor, Lyapunov exponents (LEs), bifurcation diagram, Poincare section and time domain waveform. Then, chaotic sequences are generated based on the new system and applied in a direct sequence code division multiple access (DS-CDMA) system. The randomness, correlation and anti-MAI (multiple access interference) are tested and analyzed through numeric computing and simulation. The results prove that the sequences can improve the performance of the DS-CDMA system effectively because they have good randomness, correlation and anti-MAI properties.**

Keywords-hyperchaotic system; Lyapunov exponent; chaotic sequence; DS-CDMA; anti-MAI

I. INTRODUCTION

Chaos, which was discovered by Lorenz in numerical experiments in 1963 [1], is still worthy of research today because it is sensitive to initial conditions, nonperiodic, pseudorandom and so on [2]. Chaos theory develops rapidly and is widely applied in the natural and social sciences [3]. Many new three-dimensional (3D) chaotic systems were discovered, in which the Chen [4], Lu [5] and Qi [6] systems are typical. However, the systems with simple structure and only one positive Lyapunov exponent (LE) have defects of weak security - allowing them to be easily cracked - and common correlation properties. Hence, Rossler introduced the first hyperchaotic system and defined it as a system with two or more positive LEs [7]. Hyperchaotic systems have more complex dynamic behavior because the phase space locus separate in more directions.

Many high-dimensional (the dimension≥4) hyperchaotic systems are constructed based on the existing low-dimensional chaotic systems by two ways as follows:

- By feeding back the output of the nonlinear controller into the low dimensional chaotic system equation, as used in the hyperchaotic Chen system [8].
- By coupling two low-dimensional chaotic systems together, as used in the Rossler system [9].

But, at present, some parametric identification technologies [10] have been able to crack hyperchaotic systems because the order of nonlinear product terms in these systems are low (the order generally is two). Thus, the complexity of hyperchaotic systems should be improved but the realization should not be more difficult.

In order to improve the simple methods of constructing hyperchaos and increase the order of nonlinear product terms in hyperchaotic equations, we present a new 5D hyperchaotic system, which is more complex because it has seven parameters and cubic nonlinear product terms. However, the circuit to realize the system is less complex and the structure is similar to that in [11]. Only three multipliers, two integrated operational amplifiers and a small number of resistances and capacitors are added into the circuit. In view of the limited length of this paper, we have to give up the description of the circuit.

In a CDMA system, signal multiplexing is realized by using different spread spectrum sequences to different users. Thus, the correlation property of these sequences determines the performance of the CDMA system because it affects the strength of multipath and MAI directly [12]. Traditional spread spectrum sequences, such as *m*-sequences, Gold sequences and complex sequences, have the defects of low complexity, weak security and insufficient quantity [13]. Chaotic sequences, if they are generated by a complex hyperchaotic system, are high complexity, have strong security and can generate a large number of spread spectrum sequences. Such chaotic sequences are more suitable for DS-CDMA systems than traditional sequences.

This paper is organized as follows: first, the basic dynamic behavior of the 5D new hyperchaotic system is analyzed by attractor, time domain waveform, LEs, Poincare section and bifurcation diagram. Second, a mathematical model is chosen, by which the chaotic sequences are generated, based on the analysis of the time domain waveform. Good randomness of the sequences is proved by the test standard for random and pseudorandom number generators, which is published by the National Institute of Standards and Technology (NIST). Finally, the autocorrelation and cross-correlation properties of the sequences are analyzed to prove the sequences are better than traditional hyperchaotic Lorenz sequences and *m*-sequences in anti-MAI when they are used as spread spectrum sequences in a DS-CDMA system. The result of simulation demonstrates the above conclusion.

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II. ANALYSIS OF THE NEW 5D HYPERCHAOTIC SYSTEM

A. Mathematical Model and Dissipation

First, we consider the following system:

$$\begin{cases} \dot{x}_1 = a(x_2 - x_1) + x_2 x_3 x_4 \\ \dot{x}_2 = b(x_1 + x_2) + x_5 - x_1 x_3 x_4 \\ \dot{x}_3 = -cx_2 - dx_3 - ex_4 + x_1 x_2 x_4 \\ \dot{x}_4 = -fx_4 + x_1 x_2 x_3 \\ \dot{x}_5 = -g(x_1 + x_2) \end{cases} \quad (1)$$

where a, b, c, d, e, f are positive real constants. When a=30, b=10, c=15.7, d=5, e=2.5, f=4.45 and g=38.5 the system is hyperchaotic.

The divergence of vector field of system (1) is:

$$\nabla V = \frac{\partial \dot{x}_1}{x_1} + \frac{\partial \dot{x}_2}{x_2} + \frac{\partial \dot{x}_3}{x_3} + \frac{\partial \dot{x}_4}{x_4} + \frac{\partial \dot{x}_5}{x_5} = -(a - b + d + f) = -29.45 \quad (2)$$

Note that when $\nabla V < 0$, the phase space volume of system (1) $V(t) = V(0)e^{\nabla V \cdot t}$ converge to zero exponentially and the locus will be attracted to a limited phase space, which is called attractor [14]. Thus, system (1) is a dissipative one.

B. Equilibrium Points and Stability

Five equilibrium points and the eigenvalues of Jacobian matrixes corresponding to equilibrium points are shown in Table I. In every eigenvalue set, not all the real parts of the eigenvalues are negative, that is to say, the five equilibrium points are all instable saddle points and system (1) has the possibility of being a hyperchaotic system.

C. Chaotic Attractor

We use variable step size 4-order/5-order Runge-Kutta method to solve equation set of system (1) within $t \in [0,20]$ and choose the data within $t \in [5,20]$ in order to ensure the system has been in chaotic status. The result (the attractor) is shown in Fig.1. The locus is a chaotic attractor because it is restricted in a limited phase space and the dynamic behavior looks like disorder. The time domain waveform of state variable x_1 shown in Fig.2 is obviously nonperiodic and appears random.

TABLE I. THE FIVE EQUILIBRIUM POINTS AND THE EIGENVALUES

Equilibrium Points	Eigenvalues
E_0=(0, 0, 0, 0, 0)	$\lambda_1 = -5, \lambda_2 = -36.91, \lambda_3 = 11.44,$ $\lambda_4 = -5.47, \lambda_5 = -4.45$
E_1=(2.39, —2.39, —6.85, 8.76, —143.09)	$\lambda_1 = 7.32 + 43.42i, \lambda_2 = 7.32 - 43.42i,$ $\lambda_3 = -31.85, \lambda_4 = 0.63, \lambda_5 = -12.87$
E_2=(1.56, —1.56, 10.50, —5.71, —93.33)	$\lambda_1 = 3.82 + 47.16i, \lambda_2 = 3.82 - 47.16i,$ $\lambda_3 = -29.43, \lambda_4 = -8.14, \lambda_5 = 0.48$
E_3=—E_1	$\lambda_1 = 7.32 + 43.42i, \lambda_2 = 7.32 - 43.42i,$ $\lambda_3 = -31.85, \lambda_4 = 0.63, \lambda_5 = -12.87$
E_4=—E_2	$\lambda_1 = 3.82 + 47.16i, \lambda_2 = 3.82 - 47.16i,$ $\lambda_3 = -29.43, \lambda_4 = -8.14, \lambda_5 = 0.48$

D. Poincare Section

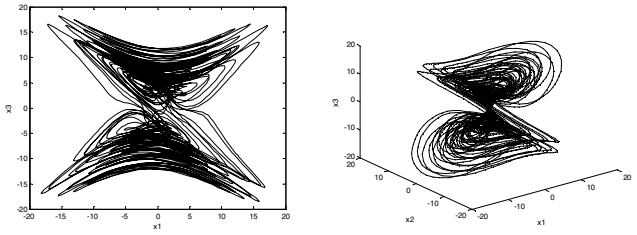
Using Poincare section to research the motion features of a multivariable autonomous system is an effective method when direct observation of the characteristic in phase space is difficult. The Poincare sections of system (1), which traverse the equilibrium point E_0, are shown in Fig.3. On the Poincare sections, masses of points are concentrated in certain areas and the leaf-like shape of the attractor can be seen. This shows that system (1) is a chaotic system.

E. Lyapunov Exponent and Bifurcation Diagram

LE, which can show the feature of most nonlinear dynamic systems, is the average exponent radiation rate of adjacent locus in phase space [14]. A system is chaotic if there is one positive LE or hyperchaotic if there are two positive LEs. Five LEs of system (1) are obtained by Gram-Schmidt reorthogonalization (GSR) algorithm [15]:

$LE_1 = 5.12, LE_2 = 0.90, LE_3 = 0, LE_4 = -10.41, LE_5 = -25.08$.

The system is hyperchaotic because there are two positive LEs.



(a) (b)

Figure 1. The chaotic attractor: (a) the attractor in x_1-x_3 plane; (b)the attractor in x_1-x_2-x_3 space.

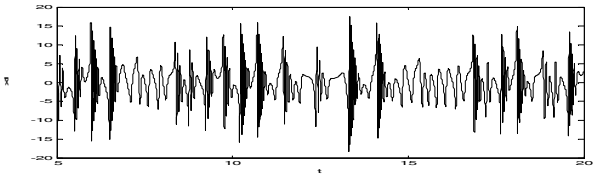
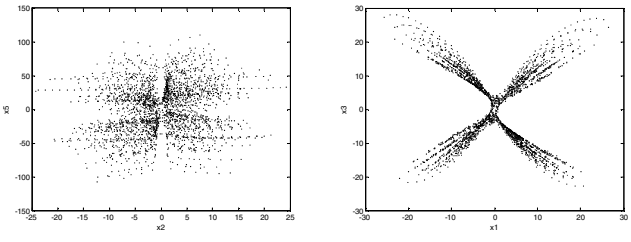


Figure 2. The time domain waveform of the state variable x_1.



(a) (b)

Figure 3. The Poincare sections: (a) x_2-x_5 plane when x_1=0; (b) x_1-x_3 plane when x_2=0

The maximum LE (LE_1=5.12) is bigger than that of most hyperchaotic systems such as hyperchaotic Lorenz system (LE_{max}= 0.3381) [16], hyperchaotic Chen system (LE_{max}= 0.5571) [17] and hyperchaotic Lü system ($LE_{max} \approx 1$) [18], which show that system (1) have more complex dynamic behavior. When a=30, b=10, c=15.7, d=5, e=2.5, f=4.45 and g=38.5, we change parameter b and draw the LE spectrums and bifurcation diagram of state variable x_1 as shown in Fig.4.

When $b \in (0.1,1.4) \cup (16.1,20)$, the locus is periodic because the LE_{max}=0; when $b \in (1.4,4.9) \cup (13.2,16.1)$, the locus is chaotic because there is one positive LE; when $b \in (4.9,13.2)$, the locus is hyperchaotic because there are two positive LEs.

Fig.5 shows the periodic attractor when b=18 and the chaotic attractor when b=1.8. Note that the chaotic attractor are simpler than hyperchaotic one as it have obvious vortexes.

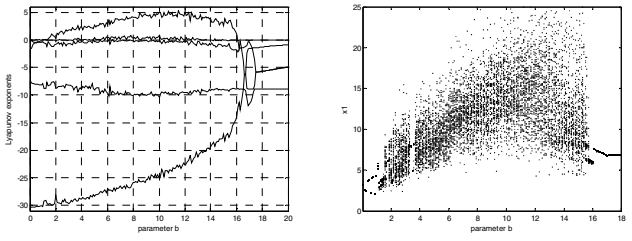
As system (1) has many parameters and the length of this paper is limited, only changes in the system state due to changes in parameter b are analyzed in detail. The effects of other parameters are given a brief summary in Table II.

III. APPLICATION IN DS-CDMA

The pseudorandomness of chaos make chaotic sequence is suitable for DS-CDMA systems as traditional sequences. However, the length and quantity of chaotic sequences are not limited because chaotic systems are sensitive to initial conditions and are nonperiodic, which is different from traditional sequences. Additionally, chaotic sequences are difficult to break, that provides protection for secure communications. Hyperchaotic systems, which are more complex than chaotic ones, can not only provide infinite spread spectrum sequences but also improve the anti-MAI ability of DS-CDMA systems because the autocorrelation of hyperchaotic sequences is like a δ function and the cross-correlation is approximately orthogonal [19].

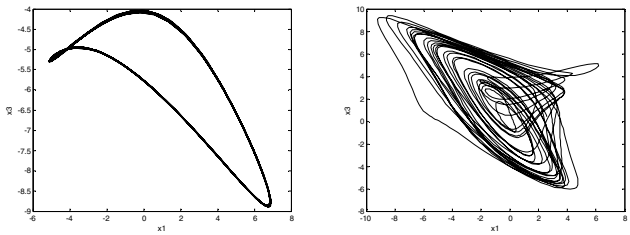
TABLE II. EFFECTS ON THE SYSTEM STATE DUE TO CHANGES IN OTHER PARAMETERS

Other parameters		System state		
Parameter	***Value range***	***Periodic***	***Chaotic***	***Hyperchaotic***
a	(20,80)	(20,23.4)	(23.4,27.9)□ (57.4,80)	(27.9,57.4)
c	(0,200)	Null	(100,200)	(0,100)
d	(0,25)	(12,1,25)	(0,2.8)∪ (9.7,12.1)	(2.8,9.7)
e	(0,10)	(9.2,10)	(7.2,9.2)	(0,7.2)
f	(0.6,25)	(16.7,25)	(0.6,2.7)∪ (11.4,16.7)	(2.7,11.4)
g	(0,200)	(135,200)	(0,2)□(67,135)	(2,67)



(a) (b)

Figure 4. The LE spectrums and the bifurcation diagram of x_1 when parameter b is changed.



(a) (b)

Figure 5. The attractor when b=18 and b=1.8: (a) the periodic attractor in x_1-x_3 plane when b=18; (b) the chaotic attractor in x_1-x_3 plane when b=1.8.

A. Generating Binary Hyperchaotic Sequences

In Fig.2, the distribution of the value of x_1 is uneven along time axis. If 0 is simply taken as the threshold of binarization, it is difficult to choose an appropriate sampling frequency because sampling slowly will result in the loss of chaotic information and while sampling quickly will result in strong local monotonicity in the loose parts of the waveform, which will worsen the randomness of the binary sequences. Therefore, choosing a suitable generating algorithm for a binary chaotic sequence is important.

Among the generating algorithms, which are summarized in [20], some are good but too complex and some are simple but the randomness of the chaotic sequences is not satisfactory due to the traditional chaotic system is simple. In other words, if the chaotic system is complex enough, the simple algorithms may be satisfactory.

System (1) is a complex hyperchaotic system, we use high sampling frequency to obtain real hyperchaotic sequences and adopt the simple algorithm presented in [21], which can overcome local monotonicity, to generate binary hyperchaotic sequences. The algorithm is:

1. Map the real sequences into interval [0, 1] by the equation

$$\hat{x}_n(t) = 10^q x_n(t) - round(10^q x_n(t)) + 0.5 \qquad (3)$$

where $n \in \{1,2,3,4,5\}$, $x_n(t)$ is the sampling value of state variable x_n , $\hat{x}_n(t)$ is the mapped value , $round(X)$ is the nearest integer to X, q is the right shift digit of $x_n(t)$ and in this paper q=3.

2. Transform decimal number $\hat{x}_n(t)$ into binary number $b_n(t)$, i.e.

$$\hat{x}_n(t) = b_n(t) = 0.b_{n1}b_{n2}..b_{nm}, b_{ni} \in \{0,1\}, i = 1,2,...,m$$

where $\{b_{ni}\}$ is the binary hyperchaotic sequences.

At sampling time t, five binary hyperchaotic sequences, whose length are m bits (m can be infinite), are obtained. Spread spectrum sequences for DS-CDMA can be constructed by any amount of any bit of any $\{b_{ni}\}$ (n can be selected from 1 to 5) at any sampling time t. So, the length and the quantity of spread spectrum sequences can be infinity. Note that $\{b_{ni}\}$ itself is not suitable to be spread spectrum sequences because the autocorrelation is not ideal. We obtain a spread spectrum sequence a_{x1} by choosing the first bit of $\{b_{1i}\}$, which is generated by state variable x_1, at different time t, i.e. $a_{x1} = \{b_{11}(t_0), b_{11}(t_1), b_{11}(t_2), ..., b_{11}(t_L)\}$. a_{x2}, a_{x3}, a_{x4} and a_{x5} can be obtained by the same way.

B. Randomness Test

Binary sequences are commonly tested by balance, runs and correlation. Traditional test of balance and runs is one-sidedness because it is based on experiment data instead of statistical analysis. NIST published fifteen statistical tests [22], which are strict because they are for cryptographic applications, to test the randomness of binary sequences. As not all the tests are suitable for spread spectrum sequences, we select eight tests to test hyperchaotic sequences. Whether the randomness is good is decided by the test result *P-value*$\in$ [0,1]. If computed *P-value* is <0.01, then conclude that the sequence is nonrandom. Otherwise, conclude that the sequence is random. Note that the nearer the *P-value* is to 1, the better the randomness is. Details of the test algorithms can be found in [22]. The result is in Table III.

In Table III, the *P-value*s of the first three columns are all >0.01, which show that the three hyperchaotic sequences have good randomness. The *P-value*s of the last column appears zero in Linear Complexity Test and Discrete Fourier Transform Test , which shows that the *m*-sequence is not random but is periodic, because it is generated by linear feedback shift register. The first three tests corresponding to traditional balance, local balance and runs tests respectively are very important to a spread spectrum sequence. Sequence a_{x1} and a_{x4} are more suitable than L_{x1} for a DS-CDMA system because the *P-value*s of the first three tests of a_{x1} and a_{x4} are higher than that of L_{x1}.

TABLE III. RANDOMNESS TEST RESULTS OF THE FOUR SEQUENCES

Test	*P-value*			
	a_{x1}	a_{x4}	L_{x1}	*m-sequence*
Frequency Test	0.9396	0.6135	0.4639	0.3501
Frequency Test within a Block	0.8496	0.3155	0.1204	0.5368
Runs Test	0.2353	0.5635	0.1021	0.6596
Test of the Longest Run of Ones in a Block	0.7433	0.3541	0.7984	0.8100
Linear Complexity Test	0.5596	0.6629	0.8979	0.0000
Approximate Entropy Test	0.7996	0.6583	0.5756	0.9581
Cumulative Sums (forward)	0.9557	0.4041	0.1198	0.6901
Cumulative Sums (backward)	0.9557	0.4041	0.1198	0.2659
Discrete Fourier Transform Test	0.6597	0.7456	0.8348	0.0000

C. Performance Analysis in a DS-CDMA System

In a DS-CDMA system, using real random sequences as address codes is the best choice because the autocorrelation function of real random sequences is like a δ function and the cross-correlation function valve is approximately zero. In a DS-CDMA system, if the channel is an AWGN channel, the number of users and bit error rate (BER) are decided by the MAI intensity which is decided by the cross-correlation of address codes.

The autocorrelation of the three sequences a_{x1}, L_{x1} and 13-order *m*-sequence are good because the autocorrelation functions are all like a δ function.

As there are so many sequence pairs, the cross-correlation of only partial sequences is analyzed. The results of numerical calculation are shown in Fig.6. We can see that the peak values (PVs) of the cross-correlation functions between a_{x1} and a_{x3} (the PV is 0.056) is lower than that between L_{x1} and L_{x3} (the PV is 0.0745) and between the two different *m*-sequences (the PV is 0.082). Note that the hyperchaotic sequence pairs are selected arbitrarily, which proves that the cross-correlation of the hyperchaotic sequences is better than that of hyperchaotic Lorenz sequences and *m*-sequences.

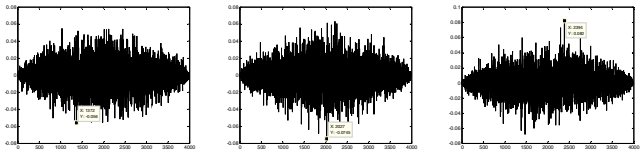
We select a_{x1}, as a typical example of the sequences generated by system (1), to be applied in a DS-CDMA system. First, a_{x1} is decomposed into K sub-sequences with length N orderly, which are assigned to users as address codes to spread the spectrum of user data. Then the data are modulated by BPSK and sent to AWGN channel. Lastly, the receiver obtains initial user data by coherent demodulation.

When the spreading gain is N and the number of users is K, the average signal-to-noise ratio (SNR) of the received signals of user i is given in [23] as follows:

$$SNR_i = \{(6N^3)^{-1}\sum_{\substack{k=1 \\ k\neq i}}^{K}[2\mu_{k,i}(0)+\mu_{k,i}(1)]+\frac{N_0}{2E_b}\}^{-1/2} \qquad (4)$$

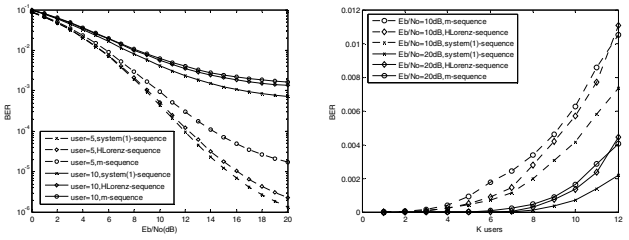
According to (4) and the formula $BER = \frac{1}{2}\mathrm{erfc}(\sqrt{SNR})$, we calculate the BER of the DS-CDMA system which uses the sub-sequences of a_{x1} as address codes. By the same way, we obtain the BER when L_{x1} and the *m*-sequence are applied in the DS-CDMA system respectively. The result is shown in Fig.7.

In Fig.7 (a), when K=5, the BER related to the two hyperchaotic sequences are similar to each other and better than the *m*-sequence. When K=10 and SNR=20dB, the BER related to L_{x1} ($\mathrm{BER} = 14\times10^{-4}$) and the *m*-sequence ($\mathrm{BER} = 16\times10^{-4}$) are almost double that related to a_{x1} ($\mathrm{BER} = 7.2\times10^{-4}$). In Fig.8 (b), when K>4 (SNR=10dB) or K>8 (SNR=20dB), the BER related to the *m*-sequence rises



(a) (b) (c)

Figure 6. The cross-correlation between partial sequences: (a) the cross-correlation between a_{x1} and a_{x3}; (b) the cross-correlation between L_{x1} and L_{x3}; (c) The cross-correlation between m-sequences generated by different generator polynomials.



(a) (b)

Figure 7. The BER of the DS-CDMA system related to a_{x1}、L_{x1} and m-sequence respectively when spreading gain N=31: (a) changes in BER due to changes in SNR when the number of users K= 5 and K=10 respectively; (b) changes in BER due to changes in K when SNR=10dB and SNR=20dB respectively.

rapidly but the acceleration is slow; the BER related to L_{x1} rise slowly but the acceleration is fast; the BER related to a_{x1} rise slowly and the acceleration is also slow. To sum up, given a certain SNR, if K is the same, then the BER related to a_{x1} is the lowest and if BER is the same, then the number of users K related to a_{x1} is the largest. As a_{x1} is a typical sequence generated by system (1), we conclude that the sequences generated by system (1) have better anti-MAI than other sequences and can improve the performance of the DS-CDMA system, which is also proved by the above experimental data.

IV. CONCLUSION

A new 5D hyperchaotic system is presented in this paper. Four equations of the system each contain a cubic product term. The system is very complex but the realizing circuit is less complex. Theoretical analysis and numerical simulation prove that the system is a new hyperchaotic system because it has all the features of a hyperchaotic system but is different from all existing hyperchaotic systems in topological structure (the number of equilibrium points and the shape of attractor). Based on the new hyperchaotic system, binary hyperchaotic sequences are generated by the selected low complexity algorithm. The randomness of the sequences is tested on the level of cryptology and the results prove that the sequences have enough good randomness to be applied in the field of cryptology. If the sequences are applied into a DS-CDMA system, they can improve the anti-MAI ability, reduce the BER and increase the number of users of the DS-CDMA system because the sequences have good randomness, autocorrelation and cross-correlation properties. The above conclusion is proved by comparing the results of numerical simulation between a typical sequence generated by the new hyperchaotic system, a traditional hyperchaotic Lorenz sequence and an m-sequence.

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